From topology-optimization brief to solver-checked release

one 24 GB graphics card · releases up to 73.7M elements · every step logged

Design brief

"four M10 bolts … a solid boss … 2 kN downward and 0.8 kN sideways …"

50 briefs: solve, clarify or refuse

user agent same tools

Typed specification

supports, loads, units, keep-in / keep-out regions

asks when a fact is missing:

design fraction 0.2 ✓

envelope 20%: 0.18

44 → 50 of 50 briefs right

with clarification under a scripted oracle

INTENT

Gate 1

13 rules

rigid body, keep-out, empty regions, device memory

error → no solve

RULES

Warm ladder



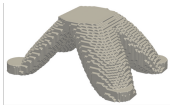
Gate 2

solver certificate

- converged
- true residual
- no memory spill
- spill monitored
- volume
- passive exact

NUMERICS

Release



mesh, report log replays without agent

14/15 released

15/15 replayed

RECORD

Release needs both gates; an agent statement cannot substitute for either

intent: engineer and clarification · admissibility: deterministic rules · numerical acceptance: solver certificate